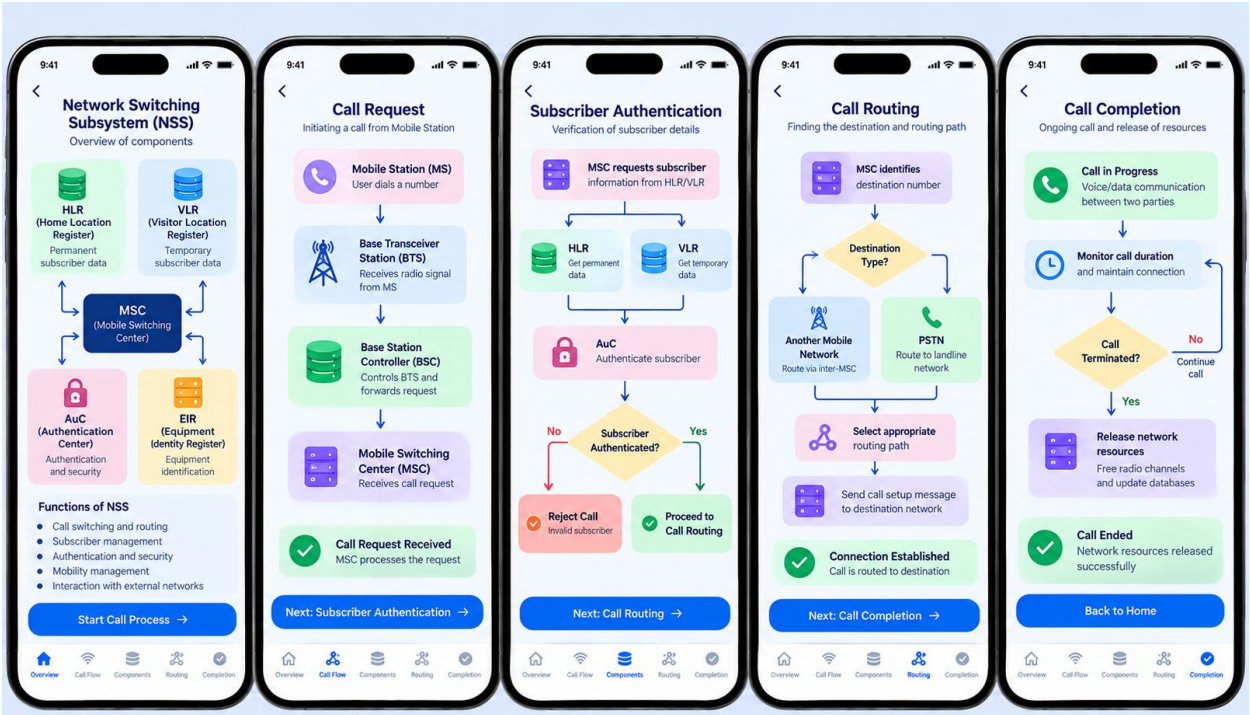

9:41
Network Switching Subsystem (NSS)
Overview of components
HLR (Home Location Register)
Permanent subscriber data
VLR (Visitor Location Register)
Temporary subscriber data
MSC (Mobile Switching Center)
AuC (Authentication Center)
Authentication and security
EIR (Equipment Identity Register)
Equipment identification
Functions of NSS
Call switching and routing
Subscriber management
Authentication and security
Mobility management
Interaction with external networks
Start Call Process →
Overview
Call Flow
Components
Routing
Completion
1. NSS Overview
9:41
Call Request
Initiating a call from Mobile Station
Mobile Station (MS)
User dials a number
Base Transceiver Station (BTS)
Receives radio signal from MS
Base Station Controller (BSC)
Controls BTS and forwards request
Mobile Switching Center (MSC)
Receives call request
Call Request Received
MSC processes the request
Next: Subscriber Authentication →
2. Call Request
9:41
Subscriber Authentication
Verification of subscriber details
MSC requests subscriber information from HLR/VLR
HLR
Get permanent data
VLR
Get temporary data
AuC
Authenticate subscriber
Subscriber Authenticated?
No
Yes
Reject Call
Invalid subscriber
Proceed to Call Routing
Next: Call Routing →
3. Subscriber Authentication
9:41
Call Routing
Finding the destination and routing path
MSC identifies destination number
Destination Type?
Another Mobile Network
Route via inter-MSC
PSTN
Route to landline network
Select appropriate routing path
Send call setup message to destination network
Connection Established
Call is routed to destination
Next: Call Completion →
4. Call Routing
9:41
Call Completion
Ongoing call and release of resources
Call in Progress
Voice/data communication between two parties
Monitor call duration
and maintain connection
Call Terminated?
No
Continue call
Yes
Release network resources
Free radio channels and update databases
Call Ended
Network resources released successfully
Back to Home
5. Call Completion